

EE4C10

Analog Circuit Design

Fundamentals

Week Q1.1

September 4th, 2025

Fabio Sebastiano

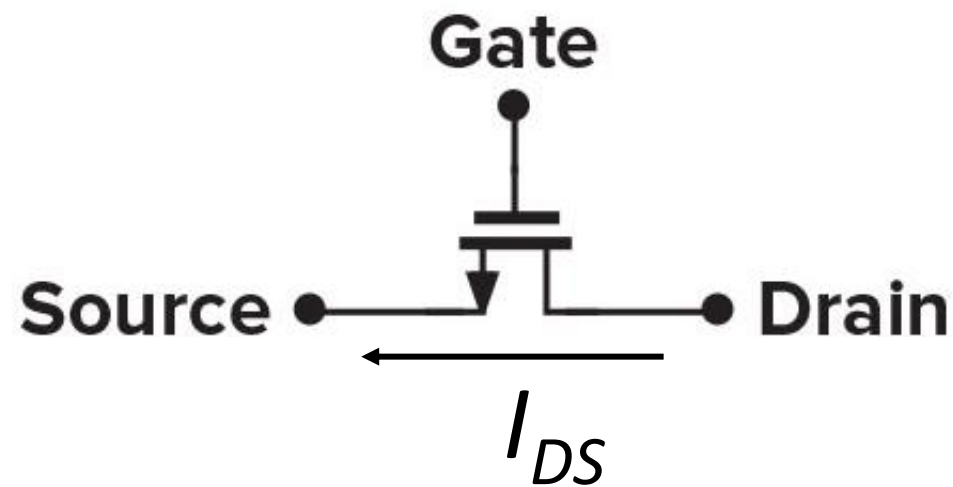
Today's goal

- Practical with LTspice
- Look at a few non-idealities of the MOS

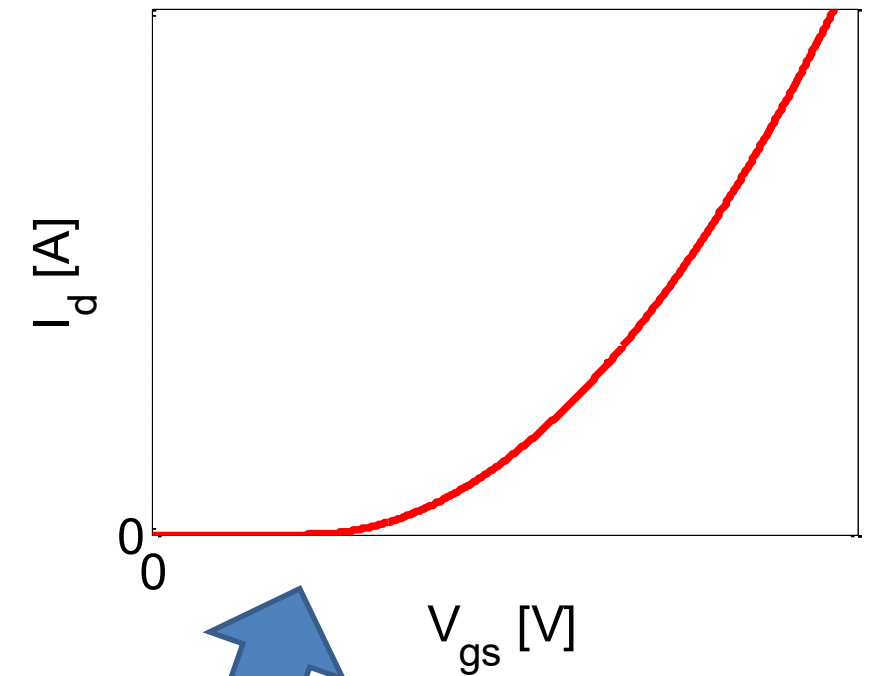
LT Spice

- We will use schematic already prepared
 - Download them from Brightspace
 - Content > Course material > LTspice files for practicals > Week 1

MOS characteristic

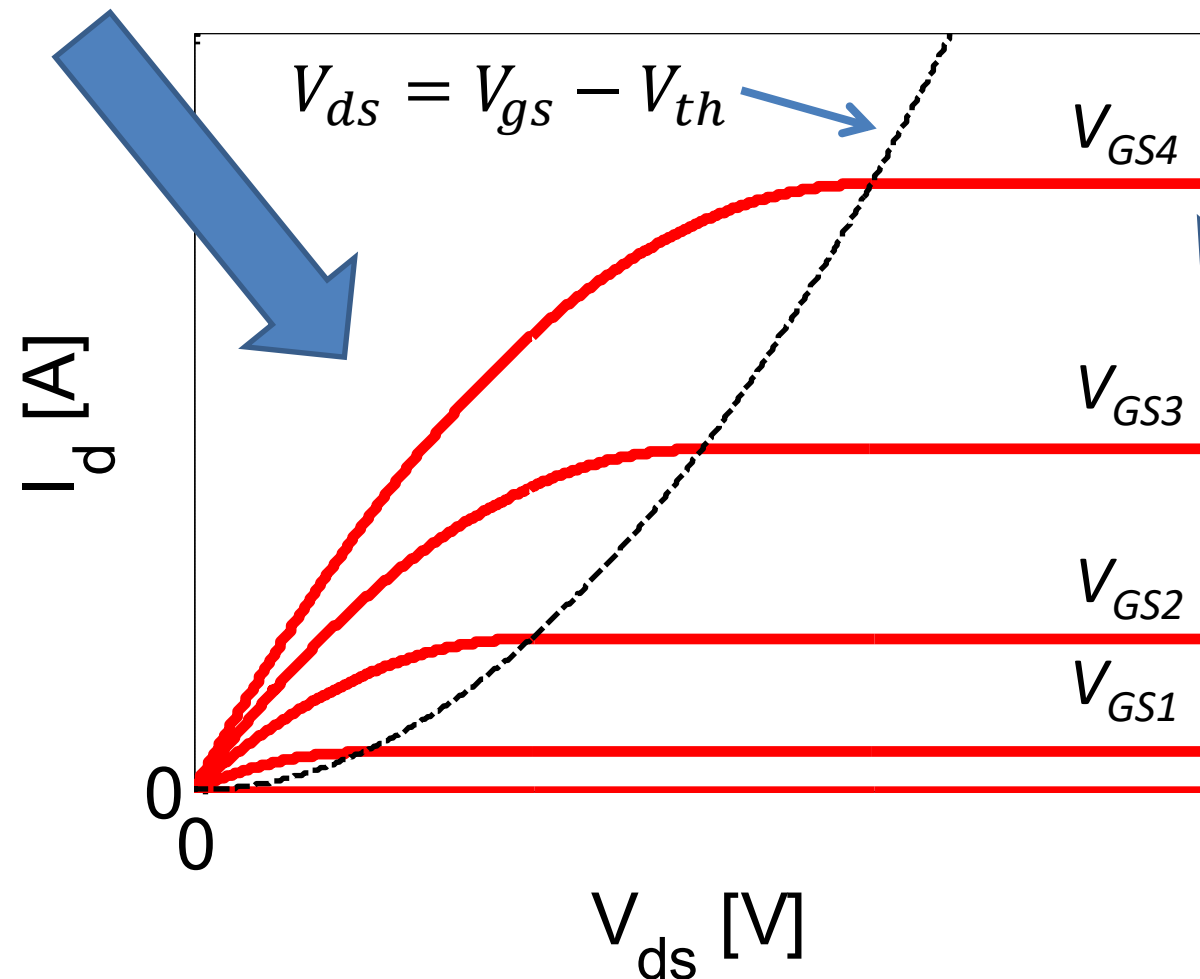


$$I_D = \mu_n C_{ox} \frac{W}{L} \left[(V_{GS} - V_{TH}) V_{DS} - \frac{1}{2} V_{DS}^2 \right]$$



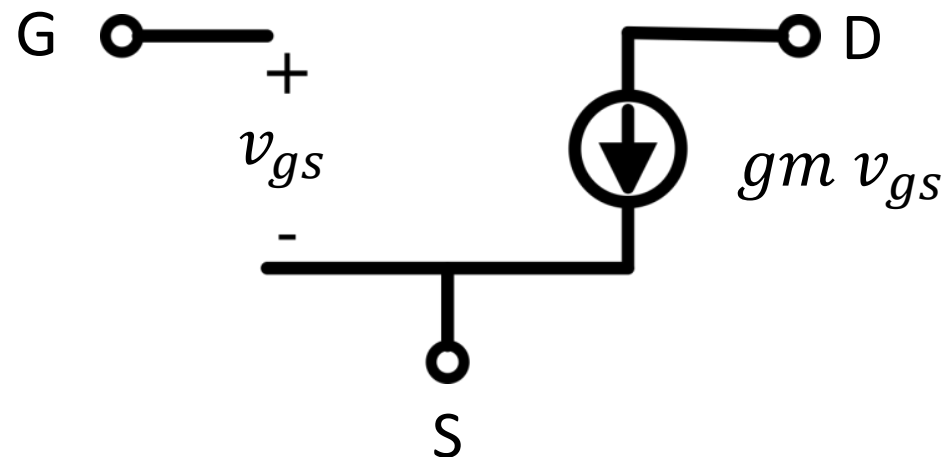
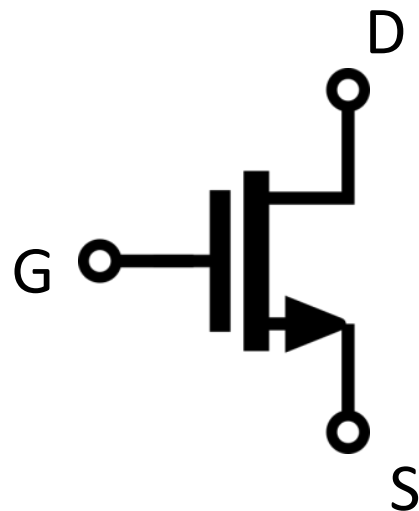
$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L'} (V_{GS} - V_{TH})^2$$

Triode
region



Saturation
region

Small-signal model – g_m



$$I_d = \frac{\beta}{2} (V_{in} - V_{th})^2$$

$$g_m = \frac{\partial i_D}{\partial v_{gs}} = \frac{\partial}{\partial v_{gs}} \frac{\beta}{2} (V_{GS} - V_{th})^2 = \beta (V_{GS} - V_{th})$$

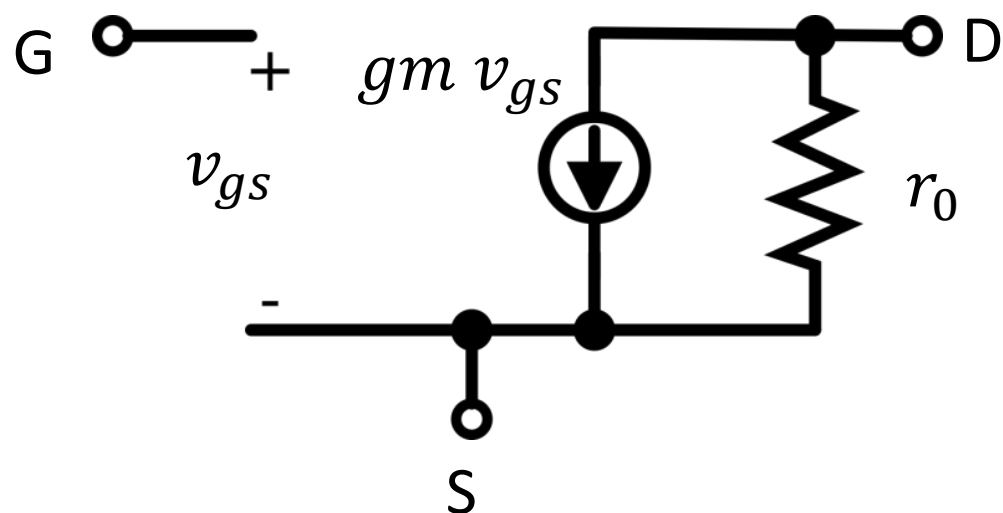
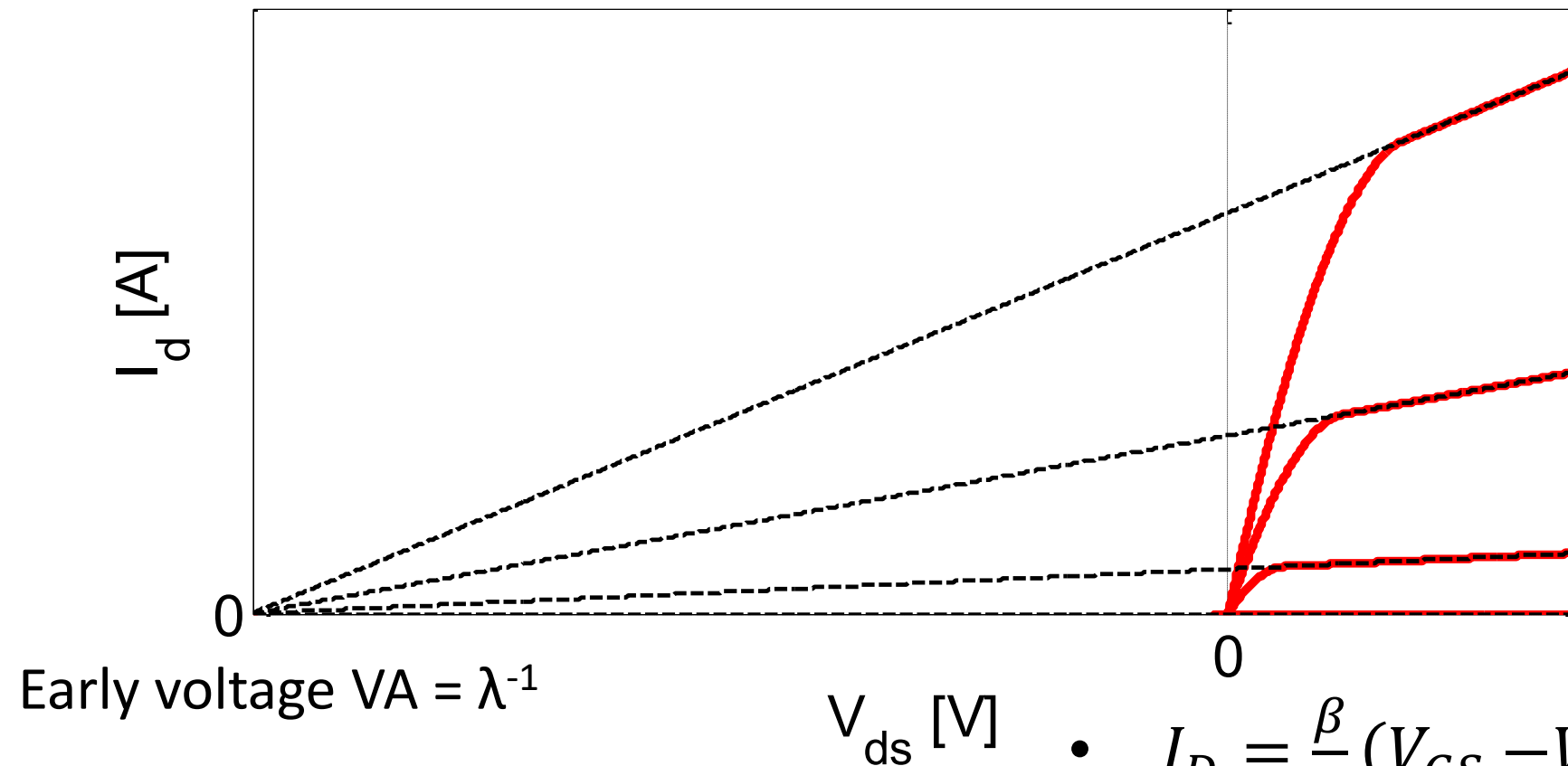
Alternative expressions:

$$= \sqrt{2\beta I_D} = \frac{2I_D}{V_{OV}}$$

- g_m : transconductance $[A/V] = [s] = [\Omega^{-1}] = [\mathcal{U}]$ (Siemens)

Valid only in saturation

Small-signal model – r_o



- $I_D = \frac{\beta}{2} (V_{GS} - V_{th})^2 (1 + \lambda V_{DS})$

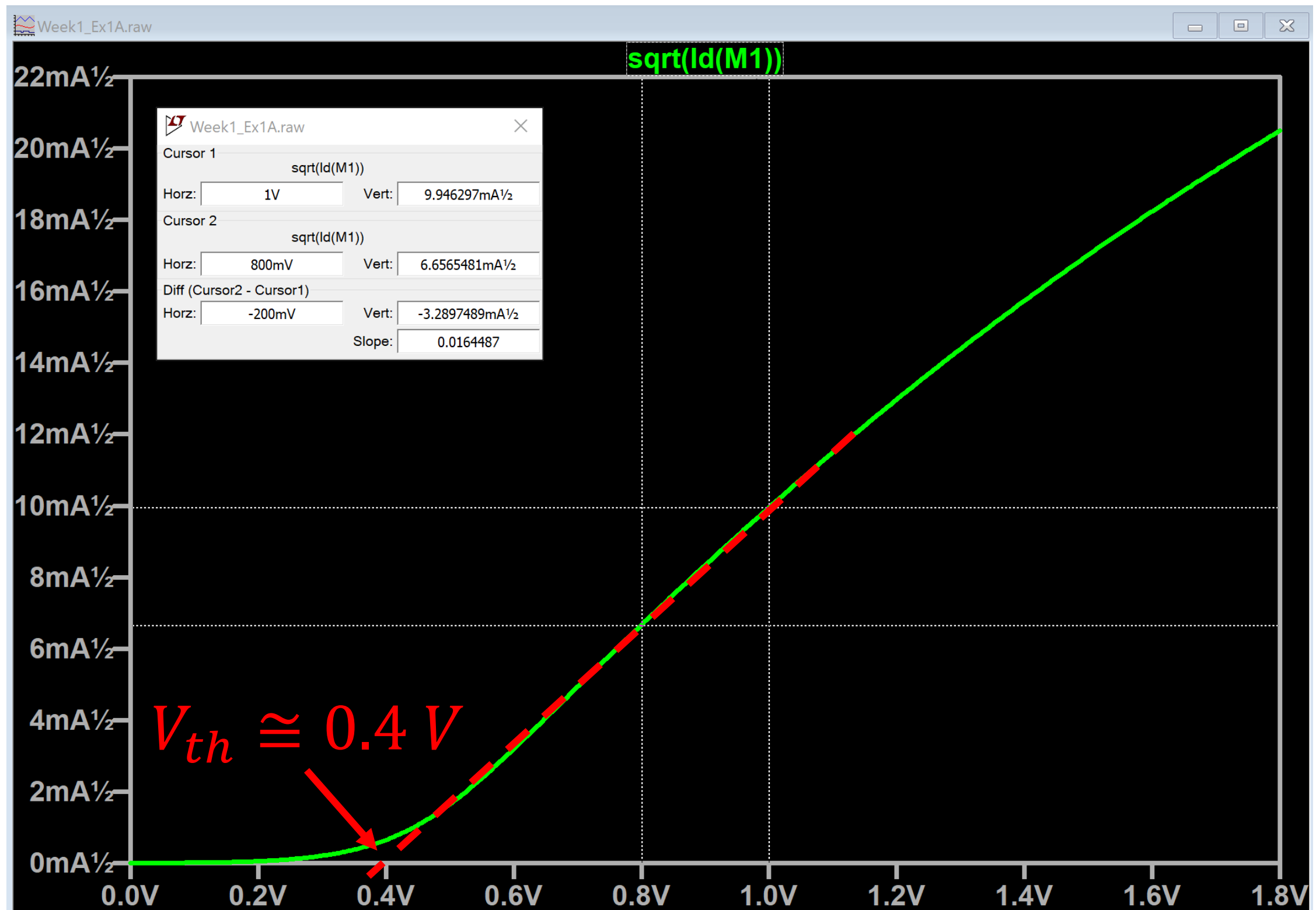
- $i_d = \frac{\partial i_D}{\partial v_{GS}} v_{gs} + \frac{\partial i_D}{\partial v_{DS}} v_{ds}$
 $= gm v_{gs} + \frac{1}{r_o} v_{ds}$

- $r_o = \left(\frac{\partial i_D}{\partial v_{DS}} \right)^{-1} =$
 $= \left(\frac{\beta}{2} (V_{GS} - V_{th})^2 \lambda \right)^{-1} \cong \frac{1}{\lambda I_D}$

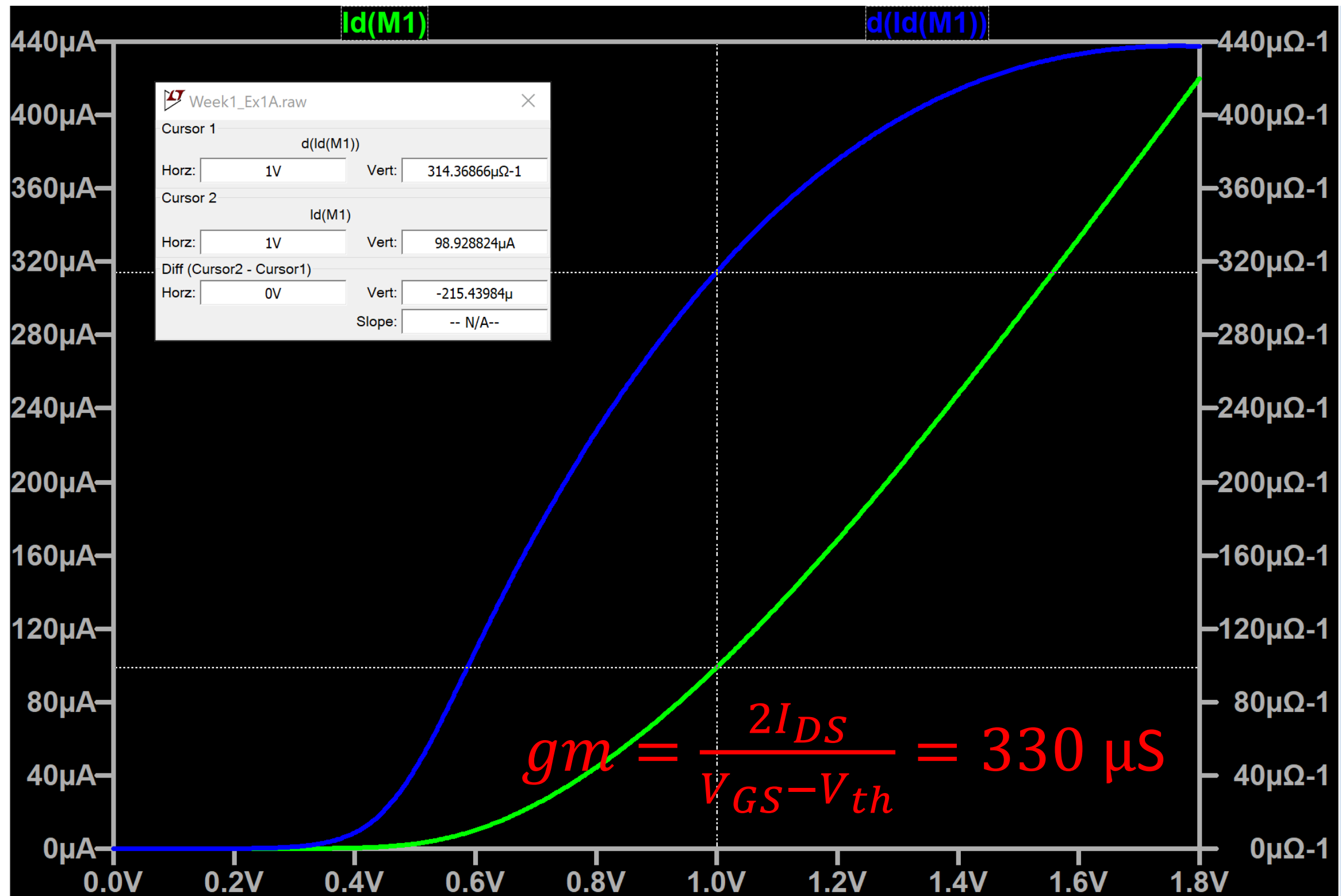
Exercise 1 – LTspice

- A. Simulate the I_{DS} vs. V_{GS} curve for an NMOS with $W=4\text{ }\mu\text{m}$, $L=2\text{ }\mu\text{m}$ and find the V_{th} . See file on Brightspace: **Week1_Ex1A.asc**
- Hint: plot $\text{sqrt}(I_{DS})$
- B. Find gm and r_o for $V_{GS}=1\text{ V}$ and $V_{DS}=1\text{ V}$:
- By computing the derivative of (I_{DS} vs. V_{GS}) and (I_{DS} vs. V_{DS})
 - Compare obtained gm with theoretical prediction
- C. Repeat point B for ($W=2\text{ }\mu\text{m}$, $L=1\text{ }\mu\text{m}$) and for ($W=0.4\text{ }\mu\text{m}$, $L=0.2\text{ }\mu\text{m}$). See: **Week1_Ex1C.asc**
- Compare the obtained values with theoretical predictions
 - Can you explain the eventual discrepancy?

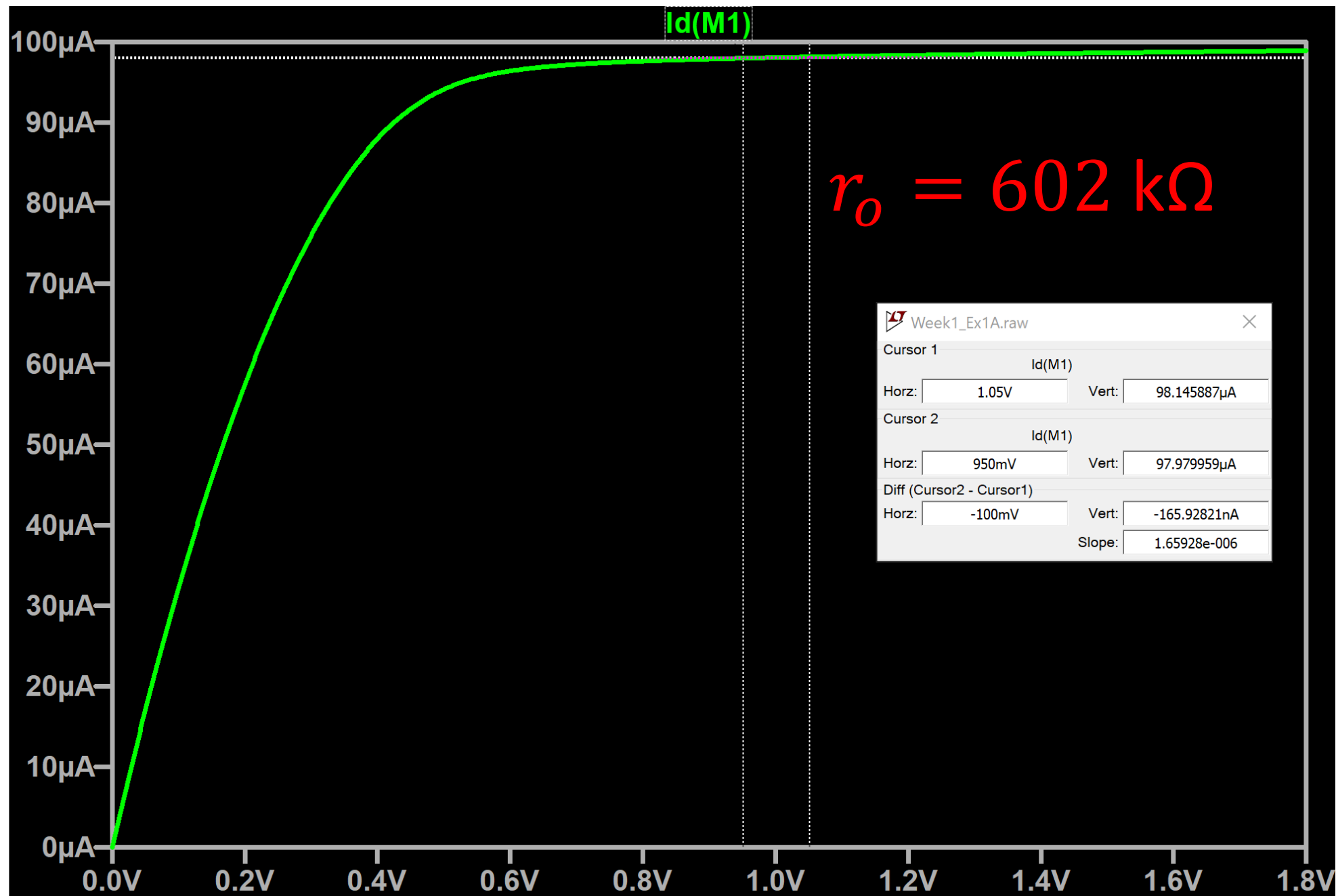
Exercise 1.A



Exercise 1.B

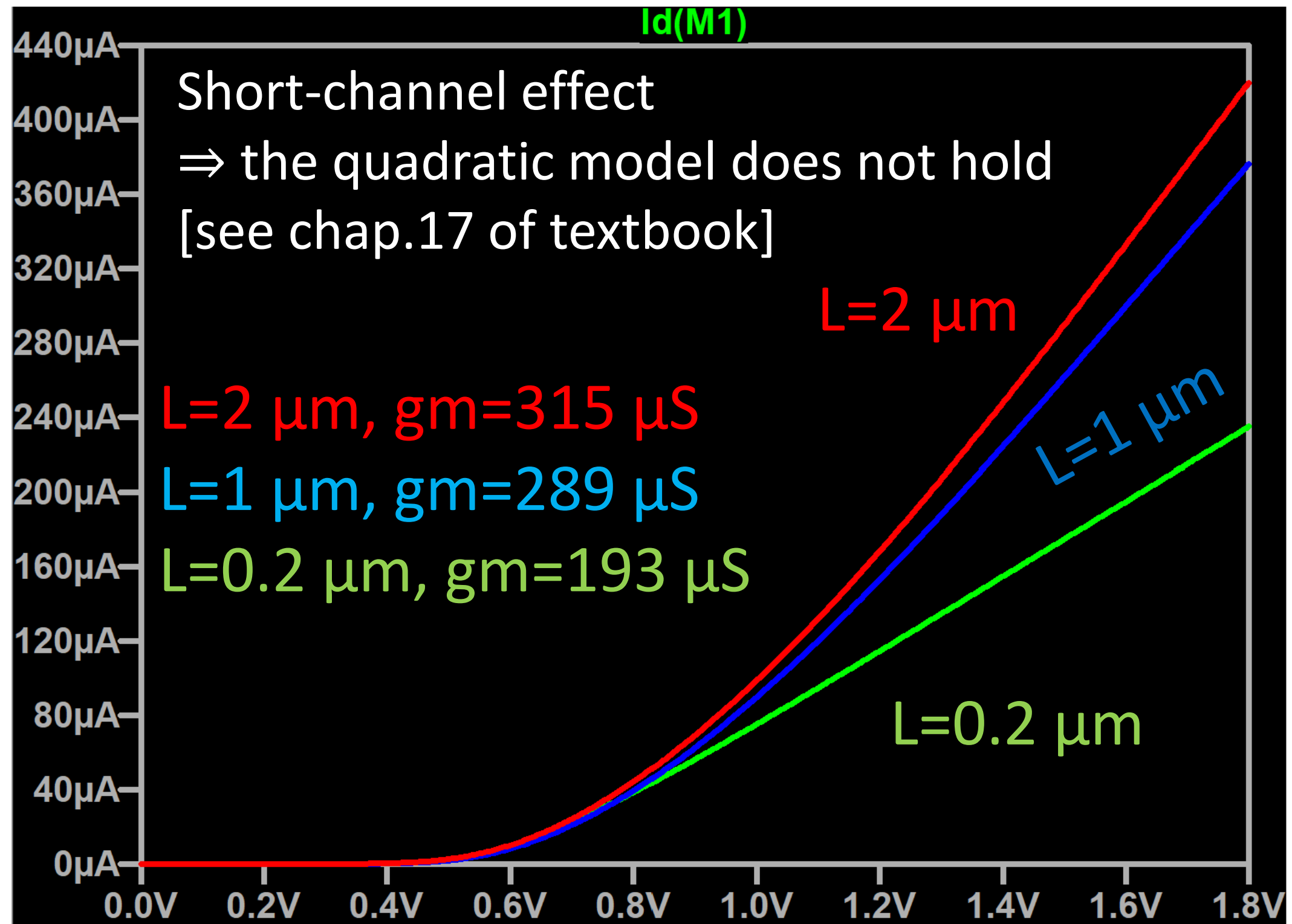


Exercise 1.B

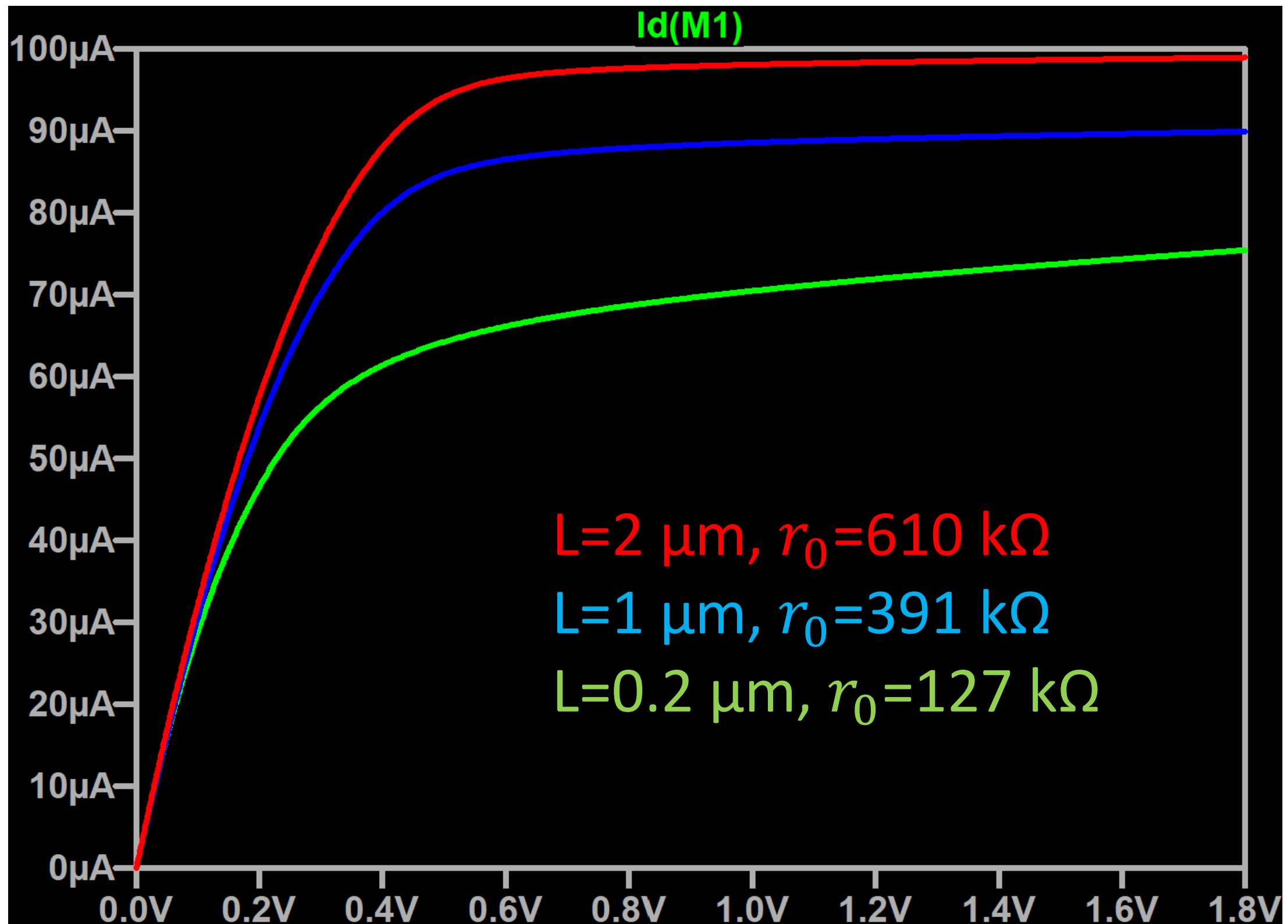


- **Note:** small-signal parameters as calculated by Spice are available by 1) performing a **.op** analysis, 2) checking the Spice report (View > Spice Error Log)

Exercise 1.C



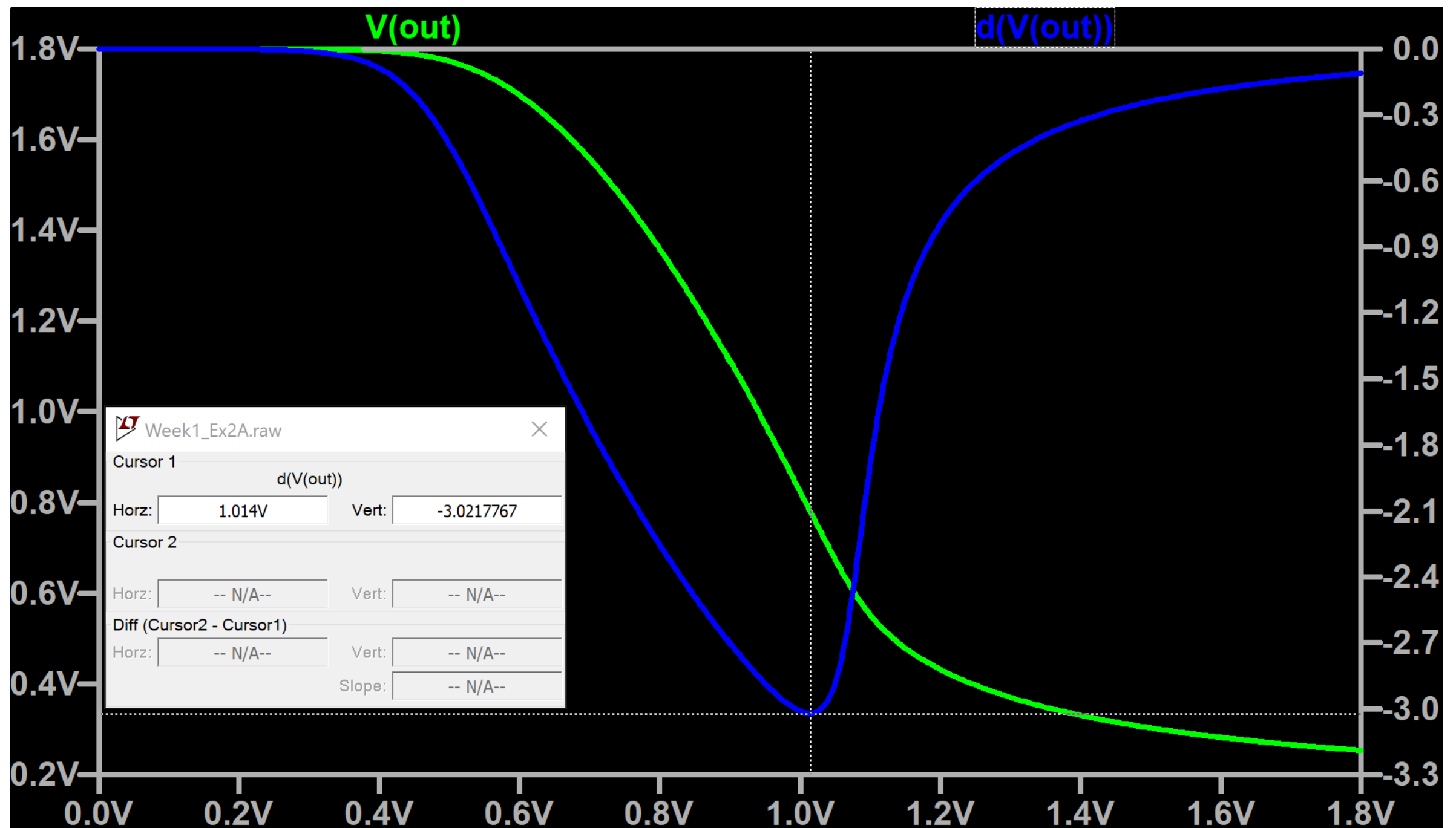
Exercise 1.C



Exercise 2 – LTspice

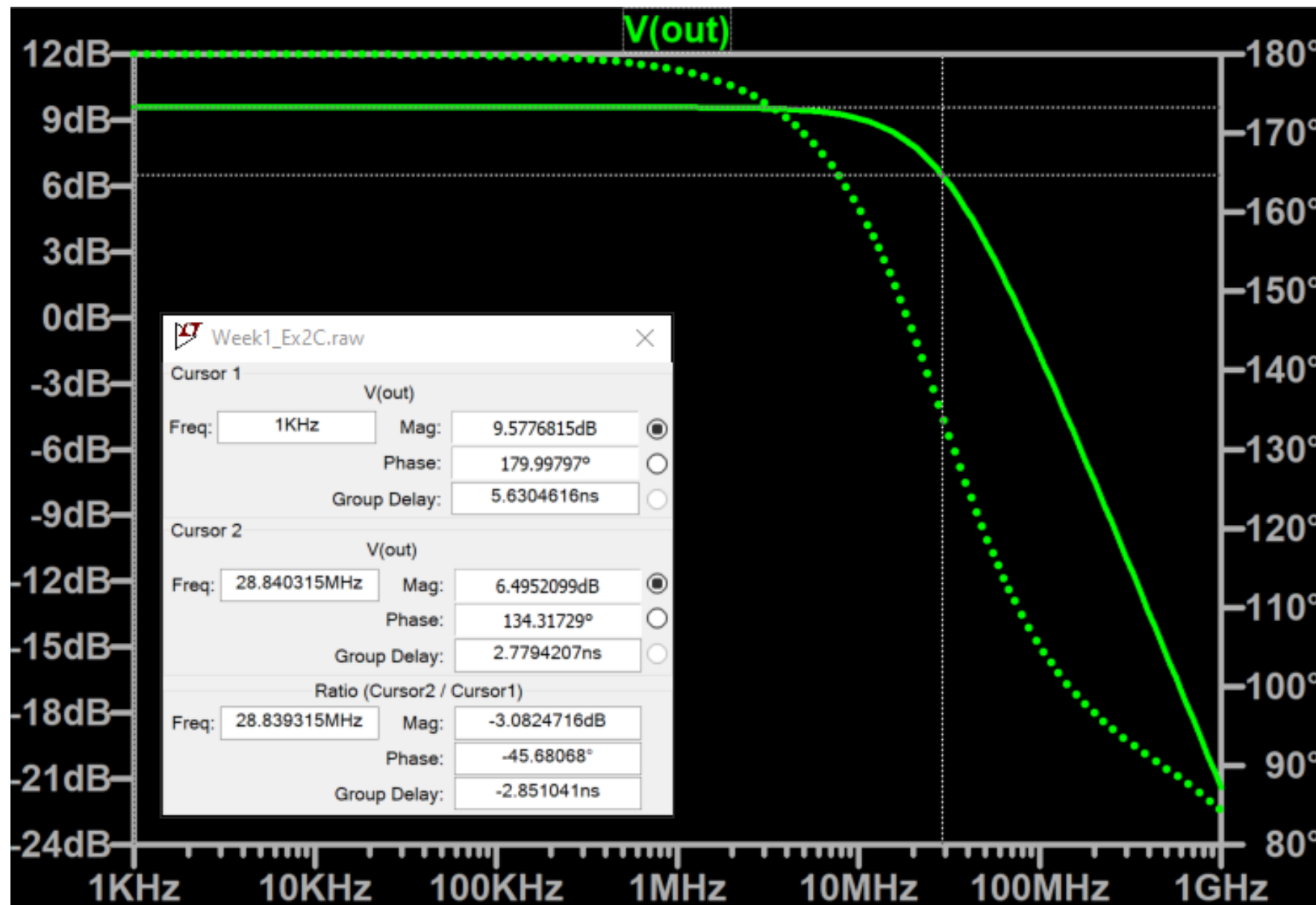
- A. Simulate the input-output characteristic of the simple amplifier in **Week1_Ex2A.asc**
 - Identify the regions of operations
- B. Find the bias conditions for maximum gain:
 - By simulation (Hint: compute derivative of I/O characteristic)
 - Compare the result to the prediction of the small-signal equivalent circuit
- C. Simulate the gain in the bias conditions as in point B but using AC analysis (see **Week1_Ex2C.asc**):
 - At a frequency $f = 1 \text{ kHz}$
 - At a frequency $f = 100 \text{ MHz}$
 - Can you explain the difference?

Exercise 2.A / 2.B



$$A_{DC} = gm \cdot R_1 = 315 \mu S \cdot 10 \text{ k}\Omega = 3$$

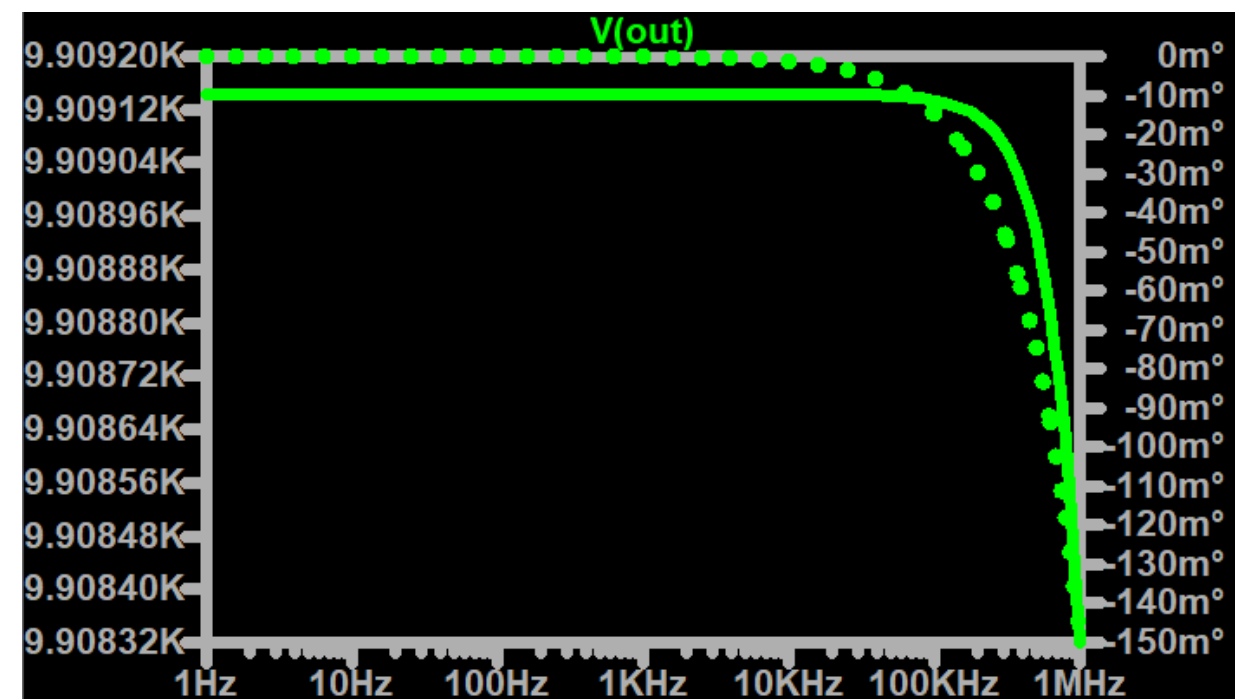
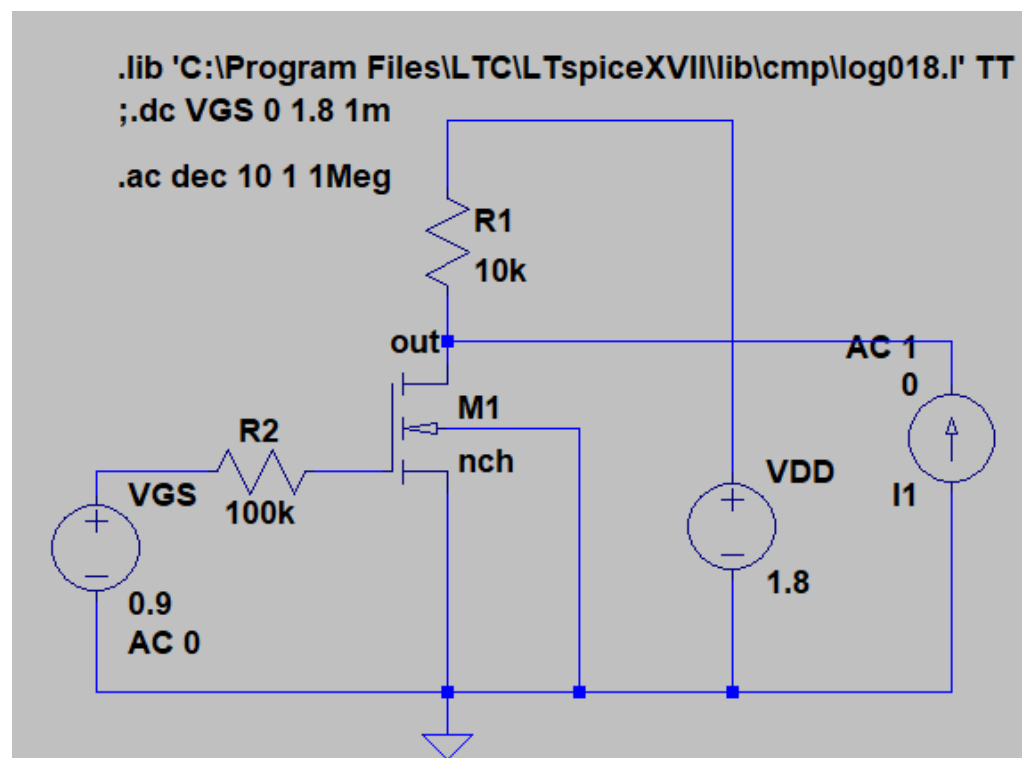
Exercise 2.C



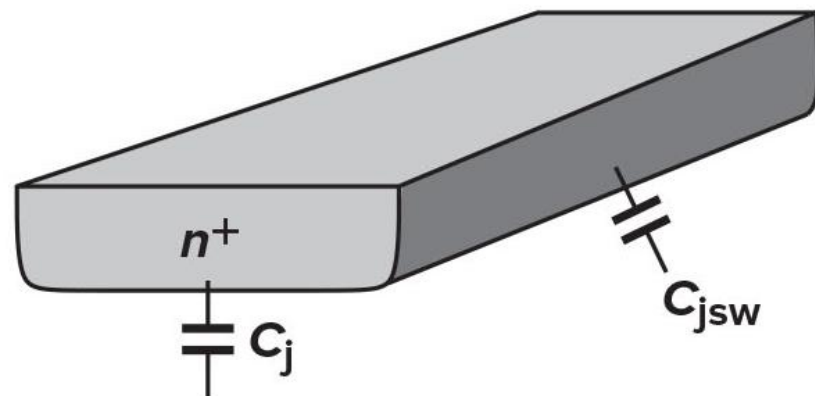
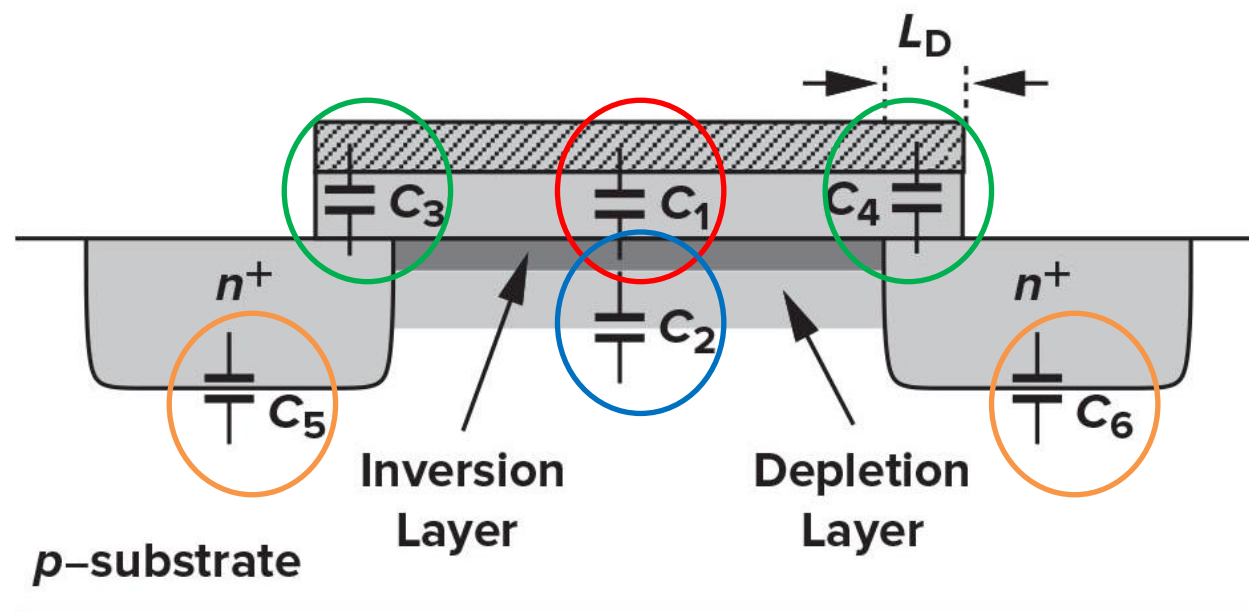
- $$f_p = \frac{1}{2\pi R_2 C} \Rightarrow C = \frac{1}{2\pi R_2 f_p} = \frac{1}{2\pi \cdot 100 \text{ k}\Omega \cdot 28.8 \text{ MHz}} = 55 \text{ fF}$$

Extra question

- How to simulate the output impedance?
 1. Properly bias the amplifier
 2. Put all AC sources to 0
 3. Add a source to stimulate the output node
 - Important: the DC value should not alter the DC bias of the circuit $\Rightarrow$ e.g. use a current source with 0 A
 4. Do an AC analysis

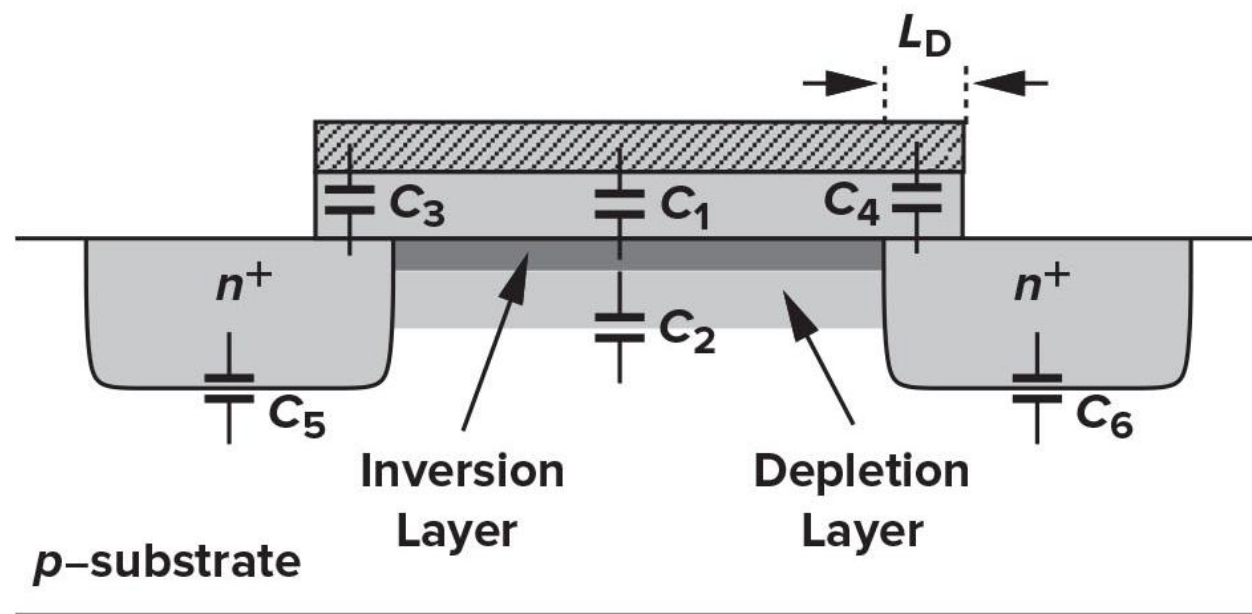


MOS Device Capacitances

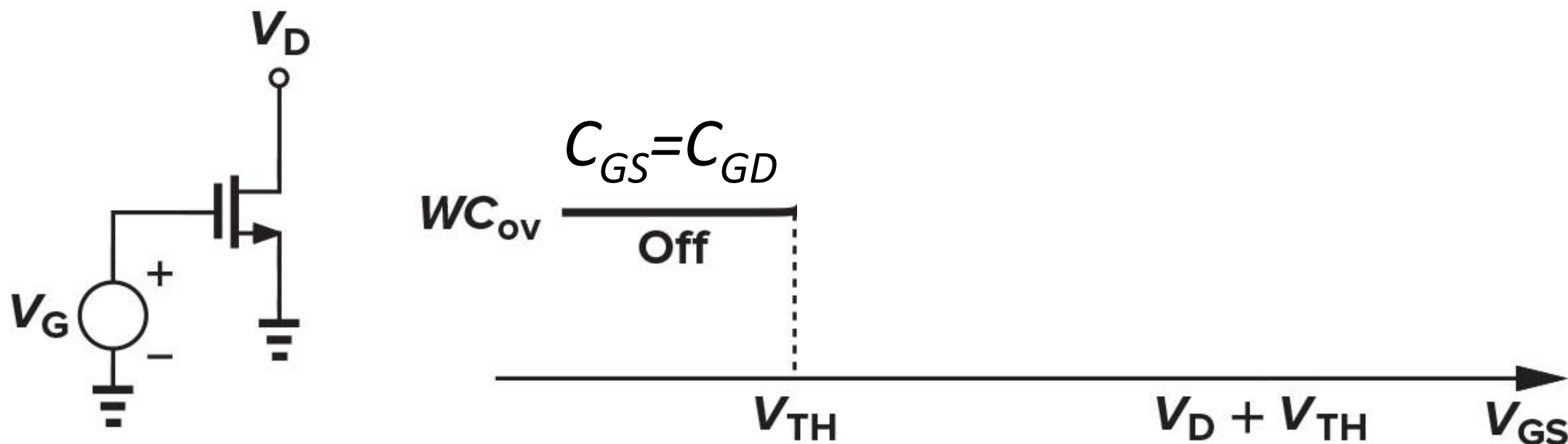


- Oxide capacitance (in triode)
 - $C_1 = WLC_{ox}$
 - $[C_{ox}] = F/m^2$
- Substrate-channel depletion cap
 - Similar to diode capacitance
- Overlap capacitances
 - $C_3 = C_4 = W \cdot C_{ov}$
 - $[C_{ov}] = F/m$
- Junction capacitance
 - Diode depletion cap
 - Depends on area **and** perimeter

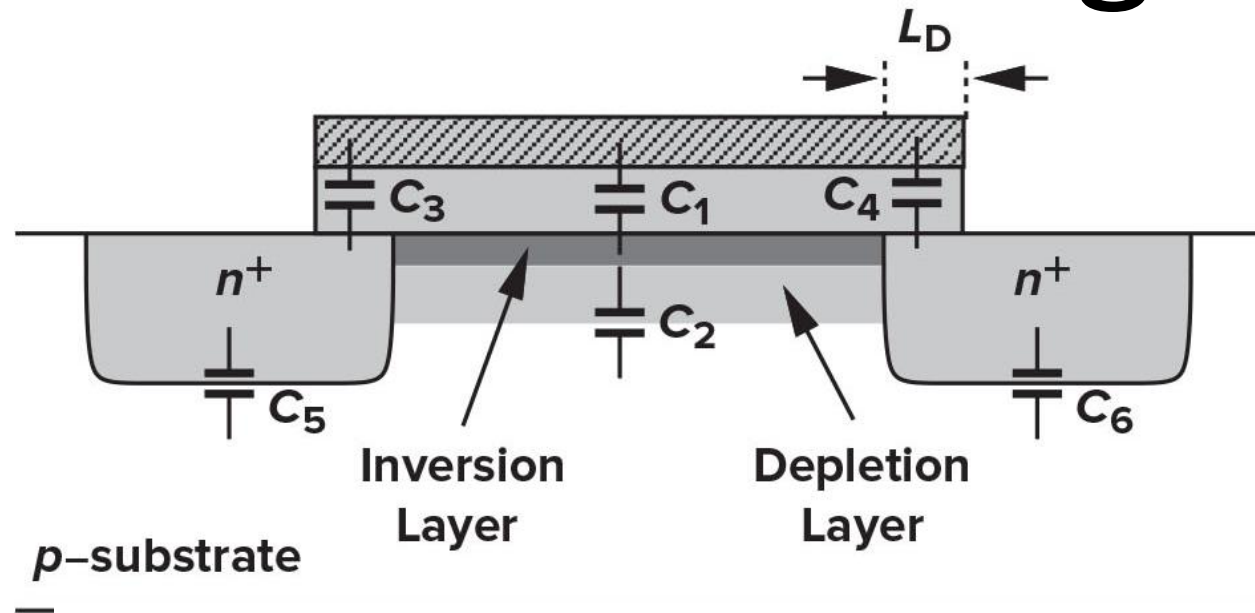
The MOS gate capacitance



- MOS is off
(better, in weak inversion)
 - No channel
 - $C_{GS} = C_{GD} \cong WC_{OV}$
 - $C_{GB} = C_1 // C_2$ negligible



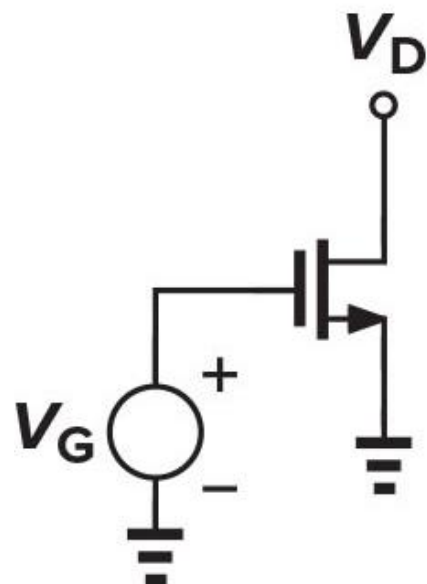
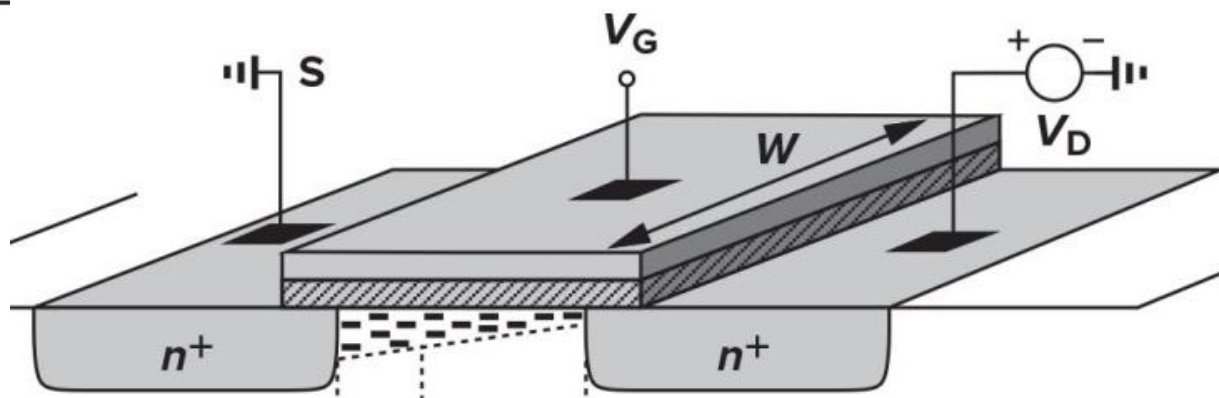
The MOS gate capacitance



- MOS is in **triode**

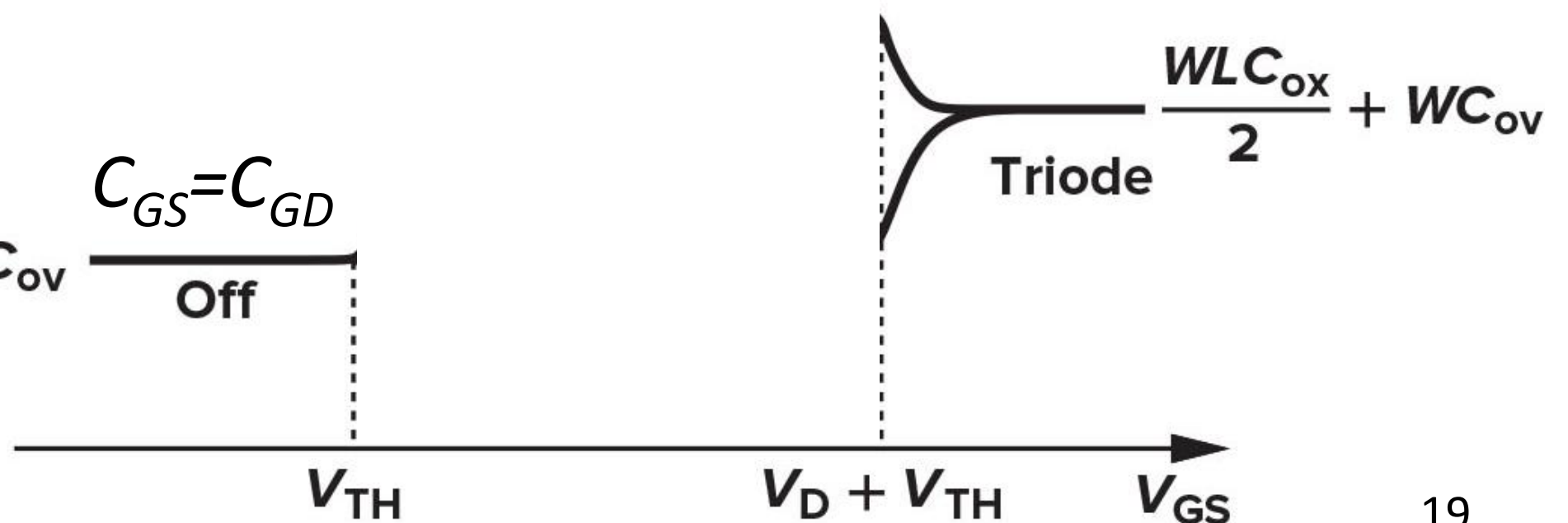
- Channel is present
- Channel shields gate from substrate

- $C_{GS} = C_{GD} \cong WC_{ov} + \frac{WLC_{ox}}{2}$

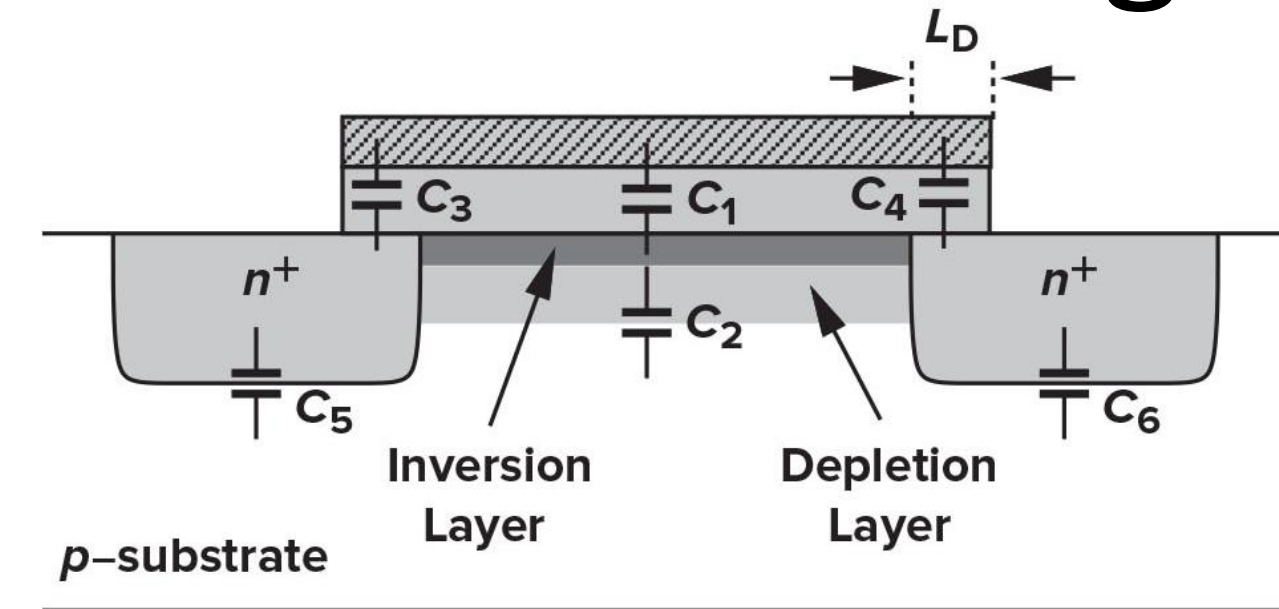


$C_{GS} = C_{GD}$

WC_{ov} Off



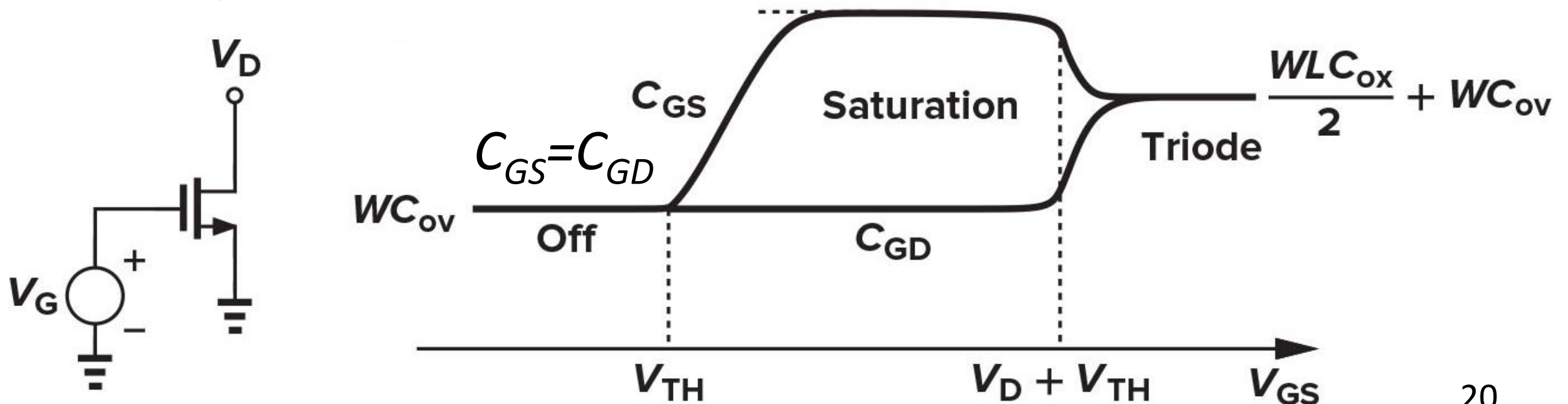
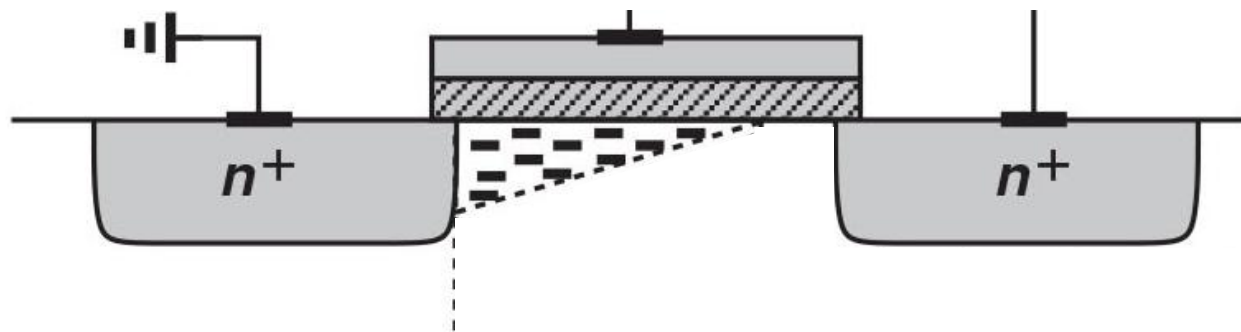
The MOS gate capacitance



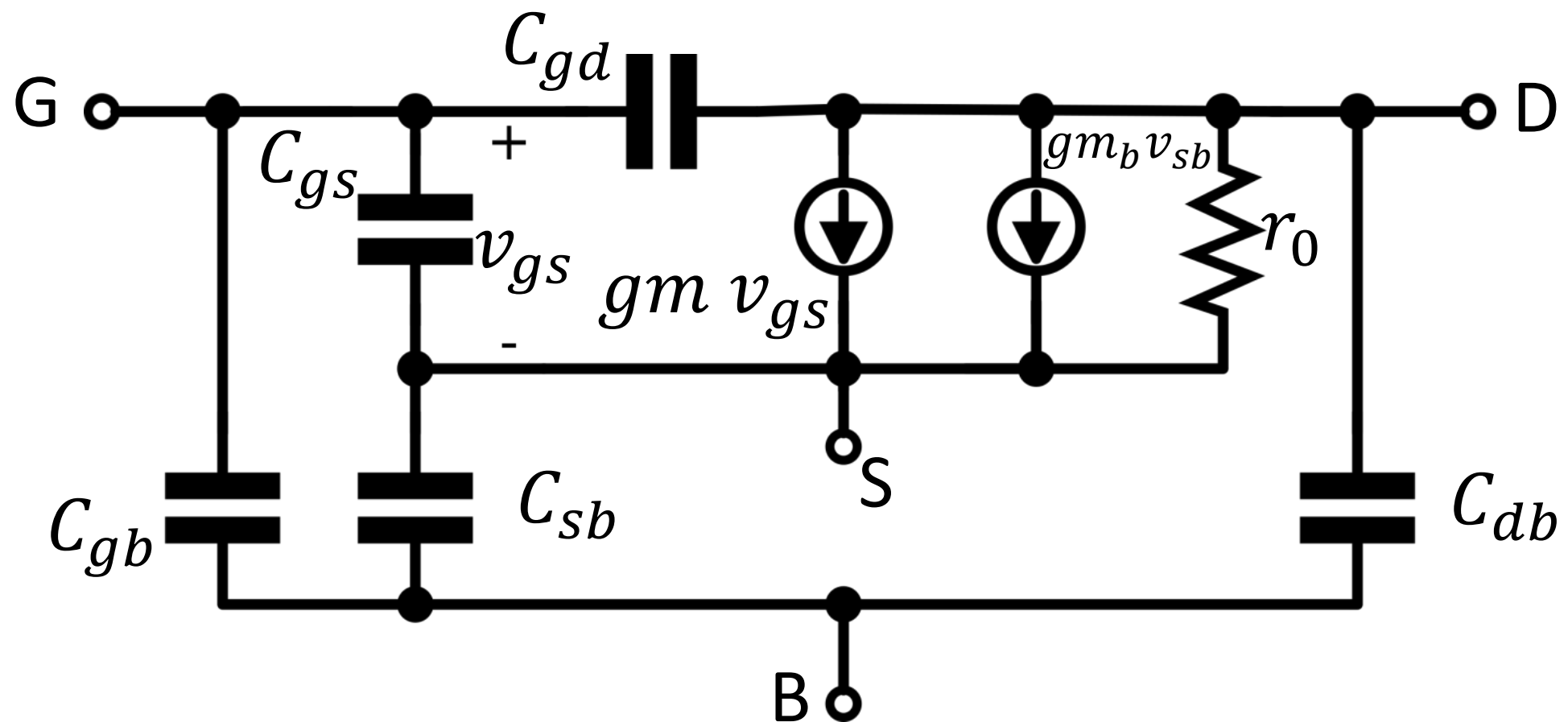
- MOS is in **saturation**

- Channel is present
- Channel shields gate from substrate

- $C_{GS} = \frac{2}{3} W L C_{ox}$, $C_{GD} = W C_{ov}$



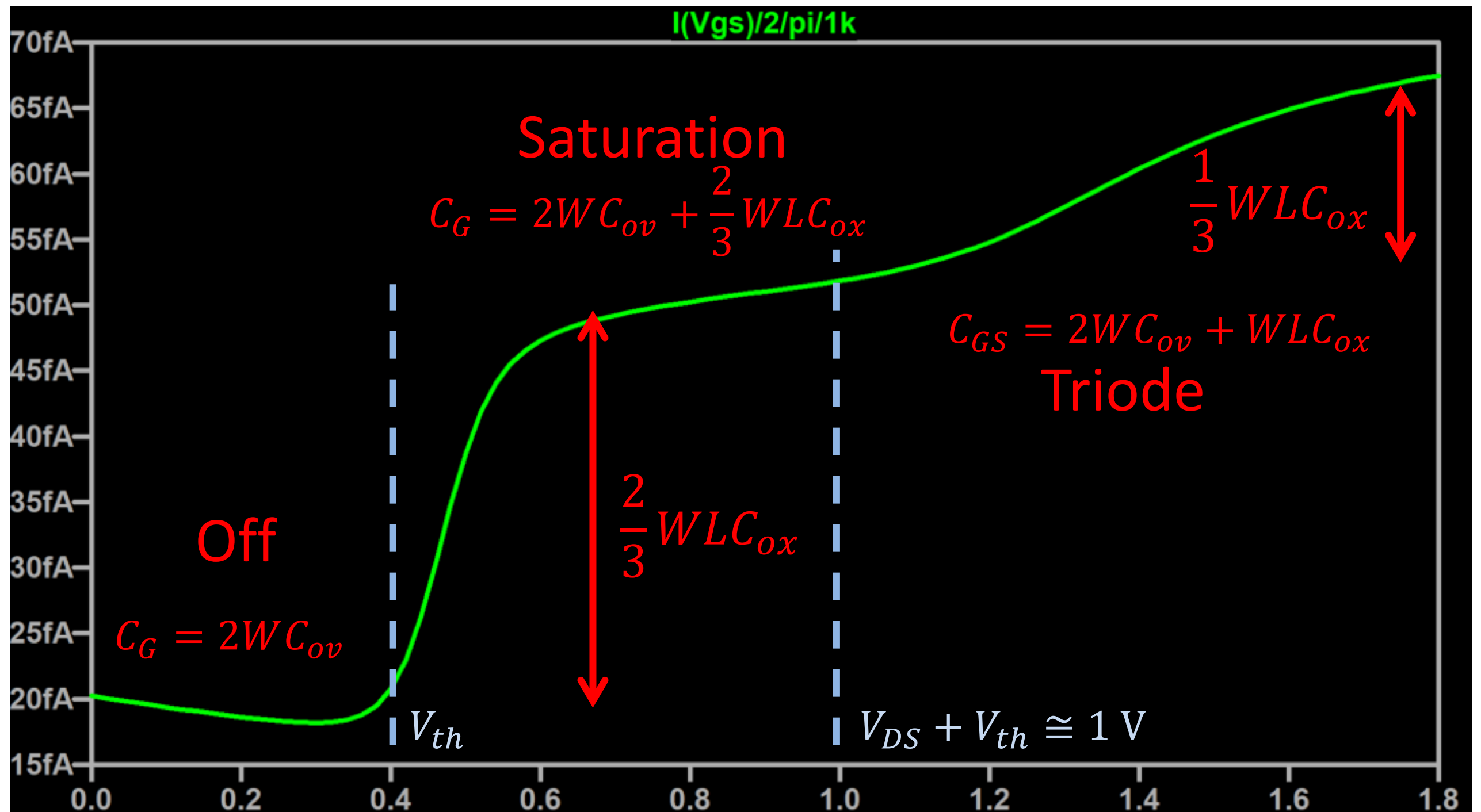
Complete MOS model



Exercise 3 – LTspice

- A. Simulate the input capacitance of an NMOS as a function of V_{GS} . Keep $4\text{ }\mu\text{m}$, $L=2\text{ }\mu\text{m}$, $V_{DS}=0.6\text{ V}$. See: **Week1_Ex3A.asc**
 - Can you explain the shape of the curve?
- B. Extract the various capacitive components of the small-signal equivalent circuit.

Exercise 3



$$WLC_{ox} \cong 50 \text{ fF} \Rightarrow C_{ox} \cong 6 \text{ fF}/\mu\text{m}^2$$

Take-home messages

- Spice simulators
 - DC and AC simulations
 - Extracting small-signal parameters, e.g. g_m and r_o
 - Simulate a simple amplifier
- Additional second-order effects
 - MOS capacitances
 - Hints to short-channel effects